# Source coding

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## Codes

**Definition 0.1** (Binary code)**.** Let $X$ be a discrete random variable. A **binary code** for $X$ assigns each value $x$ a *codeword*, a finite string of binary digits, of length $\ell(x)$.

**Definition 0.2** (Uniquely decodable code)**.** A binary code is **uniquely decodable** if every finite sequence of codewords, written one after another, can be split back into codewords in only one way.

**Definition 0.3** (Expected length)**.** The **expected length** of a binary code for $X$, where $X$ has probability mass function $p(x) \stackrel{\text{def}}{=} \mathbb{P}(X = x)$, is

$$L \stackrel{\text{def}}{=} \mathbb{E}[\ell(X)] = \sum_x p(x)\,\ell(x).$$

*Remark* 0.1 (Short codewords for common values). A code with a small expected length gives frequent values short codewords and rare values long ones, since each value's length counts in $L$ in proportion to its probability.

## The source coding theorem

**Theorem 0.1** (Source coding theorem)**.** *Let $X$ be a discrete random variable that takes finitely many values, at least two of them with positive probability, and let $H_2(X)$ be its entropy with base-2 logarithms.*

1. *Every uniquely decodable binary code for $X$ has expected length $L \geq H_2(X)$.*

2. *Some uniquely decodable binary code for $X$ has expected length $L < H_2(X) + 1$.*

3. *Coding blocks of $n$ independent copies of $X$ at a time, some uniquely decodable binary code spends, on average, fewer than $H_2(X) + 1/n$ binary digits per copy.*

*Remark* 0.2 (Where the theorem comes from). Shannon (1948) proved that entropy is the limit on how far the output of a source can be compressed. Cover and Thomas (2006, sec. 5.3 to 5.5) prove each part of Theorem 0.1. For part 2, they give each value $x$ with $p(x) > 0$ the length $\ell(x) = \lceil -\log_2 p(x) \rceil$, its surprisal in bits rounded up; the Kraft inequality (Cover and Thomas 2006, sec. 5.2) guarantees that a code with these lengths exists.

## Coding with the wrong distribution

**Theorem 0.2** (The cost of coding with the wrong distribution)**.** *Let $X$ take finitely many values, with probability mass function $p$, and let $q$ be a probability mass function, possibly different from $p$, with $q(x) > 0$ wherever $p(x) > 0$. Suppose a code gives each value $x$ with $p(x) > 0$ the length $\ell(x) = \lceil -\log_2 q(x) \rceil$. Then its expected length $L$ under $p$ satisfies*

$$H_2(p, q) \leq L < H_2(p, q) + 1,$$

*where $H_2(p, q)$ is the cross-entropy of $q$ relative to $p$, with base-2 logarithms.*

Proof

*Proof.* Every sum here runs over the values $x$ with $p(x) > 0$. Rounding up adds less than one binary digit, so for each such $x$

$$-\log_2 q(x) \leq \ell(x) < -\log_2 q(x) + 1.$$

Multiply each part by $p(x) > 0$ and sum. Since $\sum_x p(x) = 1$,

$$-\sum_x p(x) \log_2 q(x) \leq \sum_x p(x)\, \ell(x) < -\sum_x p(x) \log_2 q(x) + 1.$$

The middle sum is $L$, and the outer sums are $H_2(p, q)$. □

**Theorem 0.3** (Cross-entropy is entropy plus divergence)**.** *Let $p$ and $q$ be probability mass functions on the values $x_1, \ldots, x_n$, with $q(x) > 0$ wherever $p(x) > 0$. Write $H_2(p)$ for the entropy of $p$, $H_2(p, q)$ for the cross-entropy of $q$ relative to $p$, and $D_2(p \,\|\, q)$ for the Kullback–Leibler divergence of $q$ from $p$, all with base-2 logarithms. Then*

$$H_2(p, q) = H_2(p) + D_2(p \,\|\, q).$$

Proof

*Proof.* Every sum here runs over the values $x$ with $p(x) > 0$. For every such $x$,

$$-\log_2 q(x) = -\log_2 p(x) + \log_2 \frac{p(x)}{q(x)}.$$

Multiply by $p(x)$ and add over $x$:

$$-\sum_x p(x) \log_2 q(x) = -\sum_x p(x) \log_2 p(x) + \sum_x p(x) \log_2 \frac{p(x)}{q(x)}.$$

Reading left to right, these are $H_2(p, q)$, $H_2(p)$ and $D_2(p \,\|\, q)$. □

**Example 0.1** (Cross-entropy, entropy and divergence for a uniform code)**.** Let $p = (1/2, 1/4, 1/4)$ be the probability mass function of a three-valued random variable, and let $q = (1/3, 1/3, 1/3)$ assign $1/3$ to each of the three values. The code below computes the cross-entropy, the entropy and the divergence in bits.

```
p <- c(1 / 2, 1 / 4, 1 / 4)
q <- rep(1 / 3, 3)
cross_entropy <- -sum(p * log2(q))
shannon_entropy <- -sum(p * log2(p))
divergence <- sum(p * log2(p / q))
c(
  cross_entropy = cross_entropy,
  shannon_entropy = shannon_entropy,
  divergence = divergence
)
#>   cross_entropy shannon_entropy      divergence
#>       1.5849625       1.5000000       0.0849625
```

The cross-entropy is $H_2(p, q) = 1.585$ bits. The entropy is $H_2(p) = 1.5$ bits, and the divergence is $D_2(p \,\|\, q) = 0.085$ bits. They satisfy Theorem 0.3: $1.5 + 0.085 = 1.585$.

**Theorem 0.4** (The divergence is nonnegative)**.** *Let $p$ and $q$ be probability mass functions on the values $x_1, \ldots, x_n$, with $q(x) > 0$ wherever $p(x) > 0$. Then the Kullback–Leibler divergence satisfies*

$$D_2(p \,\|\, q) \geq 0,$$

*with* $D_2(p \,\|\, q) = 0$ *exactly when* $q = p$.

Proof

*Proof.* Every sum here runs over the values $x$ with $p(x) > 0$. For every $t > 0$, $\ln t \leq t - 1$, with equality exactly when $t = 1$. Apply this with $t = q(x)/p(x)$, and multiply by $p(x) > 0$:

$$p(x) \ln \frac{q(x)}{p(x)} \leq q(x) - p(x),$$

with equality exactly when $q(x) = p(x)$. Add over $x$. Since $\sum_x p(x) = 1$ and $\sum_x q(x) \leq 1$,

$$\sum_x p(x) \ln \frac{q(x)}{p(x)} \leq \sum_x q(x) - 1 \leq 0.$$

The left side is $-\ln 2 \cdot D_2(p \,\|\, q)$, so $D_2(p \,\|\, q) \geq 0$.
If $D_2(p \,\|\, q) = 0$, both relations hold with equality. The first forces $q(x) = p(x)$ at every $x$ with $p(x) > 0$. The second then gives $\sum_x q(x) = 1$, so $q$ has no mass where $p$ has none, and $q = p$. Conversely, $q = p$ gives $\log_2(p(x)/q(x)) = 0$ for every $x$. □

**Example 0.2** (Divergence of a biased coin from a fair coin)**.**
Let $p = (0.9, 0.1)$ be the probability mass function of a biased coin, and let $q = (0.5, 0.5)$ be a fair coin. The code below computes the divergence both ways, in bits.

```
p <- c(0.9, 0.1)
q <- c(0.5, 0.5)
c(
  p_from_q = sum(p * log2(p / q)),
  q_from_p = sum(q * log2(q / p)),
  p_from_p = sum(p * log2(p / p))
)
#>  p_from_q  q_from_p  p_from_p
#> 0.5310044 0.7369656 0.0000000
```

Both divergences between the two coins are positive, as Theorem 0.4 requires, and they differ, so swapping $p$ and $q$ changes the divergence. The divergence of $p$ from itself is 0.

*Remark* 0.3 (Cross-entropy is the cost of the wrong code). Theorem 0.3 splits the cross-entropy into the entropy and the divergence. By Theorem 0.1, no uniquely decodable code averages fewer than $H_2(p)$ bits. By Theorem 0.2, a code built for $q$ costs about $D_2(p \,\|\, q)$ bits per value more than that. By Theorem 0.4, the excess vanishes only when the code was built for $p$ (Cover and Thomas 2006, sec. 2.6; Yeung 2008, sec. 2.5). Cover and Thomas (2006, sec. 5.4) state Theorem 0.2 in this form, with $H_2(p) + D_2(p \,\|\, q)$ in place of $H_2(p, q)$.

*Remark* 0.4 (Why models are fit by minimizing cross-entropy). Fitting a classifier by minimizing cross-entropy chooses the model's distribution $q$ so that a code built for $q$ would spend as few bits as possible on the observed labels. By Remark 0.3, that pushes $q$ toward the distribution the labels came from.
For a single labeled example, let $e$ be the distribution that puts all its mass on the observed label. Then $H_2(e) = 0$ (as for a coin that always lands heads), so the example's cross-entropy loss $H_2(e, q)$ equals $D_2(e \,\|\, q)$: every bit the loss reports is a wasted one.

## References


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Yeung, Raymond W. 2008. *Information Theory and Network Coding*. Springer. https://doi.org/10.1007/978-0-387-79234-7.